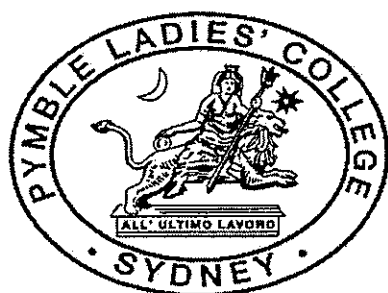


Mr Antonio
Mrs Collett
Mrs Kerr
Ms Lau
Mrs Soutar

Name:

Teacher:



2011
TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided
- All necessary working should be shown in every question
- Start each question in a new booklet
- Marks may be deducted for careless or untidy work

Total Marks – 84

- Attempt Questions 1-7
- All questions are of equal value

Mark	/84
Rank	/
Highest Mark	/84

Question 1. (12 Marks) Use a **SEPARATE** Writing Booklet.

Marks

- (a) Find $\int \frac{2}{\sqrt{36-x^2}} dx$. **1**
- (b) Differentiate $\tan^{-1}(\ln x)$ with respect to x . **1**
- (c) Find the coordinates of the point P that divides the interval joining A (-3, 8) and B (7, -3) internally in the ratio 2 : 3. **2**
- (d) Solve $\frac{4}{x-2} \leq 2$. **3**
- (e) The acute angle between the lines $2x - y = 4$ and $y = mx + 3$ is 45° . Find the two possible values of m . **2**
- (f) Use the substitution $u = x - 3$ to evaluate $\int_4^5 \frac{x}{\sqrt{x-3}} dx$. **3**

Question 2 (12 Marks) Use a **SEPARATE** Writing Booklet.

Marks

- (a) Let $f(x) = 3 \sin^{-1} 2x$. 2

Sketch the graph of $y = f(x)$, clearly indicating the endpoints for the domain and the range.

- (b) (i) Differentiate $x \cos^2 x$ with respect to x . 2

- (ii) Hence, or otherwise, find $\int x \sin 2x \, dx$. 2

- (c) The polynomial $P(x) = x^3 + ax^2 - 2x + b$ has $(x + 1)$ as a factor. $P(x)$ has a remainder of 4 when divided by $(x - 3)$. 3

Find the values of a and b .

- (d) The function $f(x) = x - e^{-2x}$ has one root between $x = 0$ and $x = 1$. 3

Use one application of Newton's method, starting at $x = 0.3$, to find another approximation for this root.

Write your answer correct to 2 decimal places.

Question 3 (12 Marks) Use a **SEPARATE** Writing Booklet.

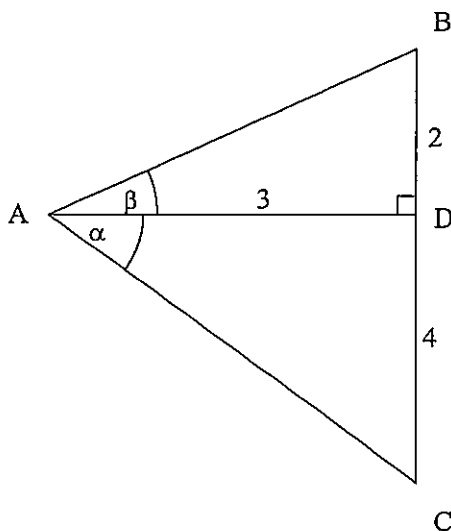
Marks

(a) Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 + 1}{2 - 7x - 5x^2}$. **2**

(b) (i) Express $\sqrt{3} \cos \theta - \sin \theta$ in the form $r \cos(\theta + \alpha)$, where $r > 0$, and $0 < \alpha < \frac{\pi}{2}$, giving r and α as exact values. **2**

(ii) Solve $\sqrt{3} \cos \theta - \sin \theta = -1$, for $0 \leq \theta \leq 2\pi$, giving your answers as exact values. **2**

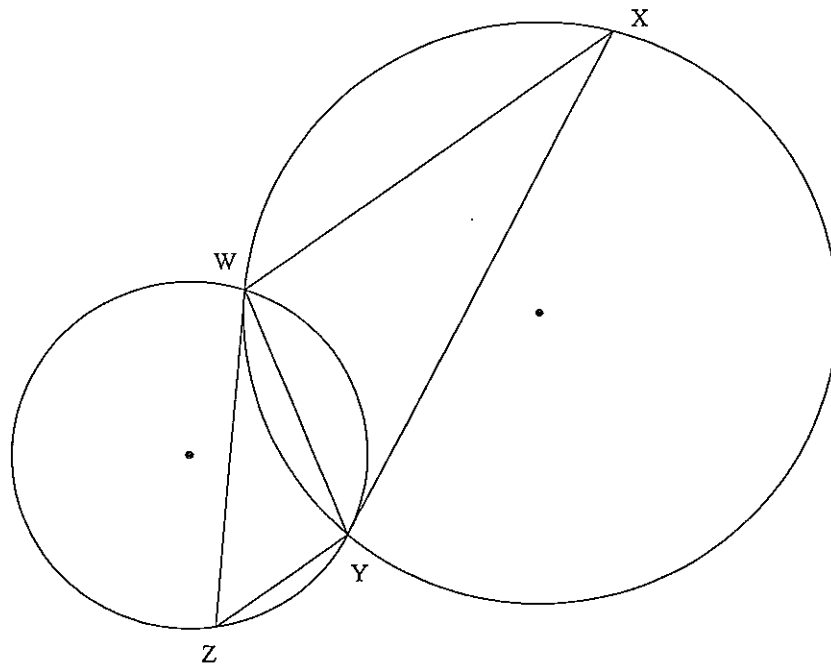
(c) In the diagram below, AD is perpendicular to BC.
CD = 4, BD = 2 and AD = 3. $\angle CAD = \alpha$ and $\angle BAD = \beta$. **3**



Find the exact value of $\sin(\alpha - \beta)$.

- (d) WZ and XY are tangents to the circles WXY and WYZ respectively. The circles share two common points, W and Y. Copy or trace the diagram into your Writing Booklet.

3



Prove that $WX \parallel YZ$.

Question 4 (12 Marks) Use a **SEPARATE** Writing Booklet.

Marks

(a) Find the exact value of $\int_0^{\ln 3} \frac{e^x}{e^x + 9} dx$. 2

(b) Find the constant term in the expansion $\left(2x - \frac{1}{x}\right)^6$. 2

(c) Prove by induction that $4p + 3p^2 + 2p^3$ is divisible by 3 for $p = 1, 2, 3, \dots$ 3

(d) The temperature ($T^\circ\text{C}$) of steel, after it has been removed from a hot furnace, after t minutes, satisfies the differential equation:

$$\frac{dT}{dt} = k(T - 22) \quad \text{where } k \text{ is a constant.}$$

Initially, the temperature (T) of the steel is 100°C and when $t = 15$ minutes, $T = 70^\circ\text{C}$.

(i) Use this information to find the exact values of A and k . 2

(ii) Hence find the value of t when $T = 40^\circ\text{C}$ to the nearest minute. 1

(e) Show that: 2

$$\sqrt{\frac{1 + \sin 2\theta}{1 - \sin 2\theta}} = \frac{1 + \tan \theta}{1 - \tan \theta}.$$

Question 5 (12 marks) Use a **SEPARATE** Writing Booklet.

Marks

- (a) The speed V cm/s of a particle moving along the x – axis is given by

$$V^2 = 27 + 18x - 9x^2 \text{ where } x \text{ is in cm.}$$

- (i) Prove that the motion is Simple Harmonic. Find the period and amplitude of the motion. 3

- (ii) Find the acceleration of the particle when it is 1cm away from the centre of motion. 1

- (b) A stone is projected upwards from the edge of a cliff with a speed of 30m/s. It hits an object 120 m horizontally from the edge and 35 m vertically below it.

Assume that t seconds after the release, the position of the stone is given by

$$x = 30t \cos \alpha \text{ and } y = -5t^2 + 30t \sin \alpha .$$

- (i) Find α , the angle of projection, to the nearest minute. 3

- (ii) Find the time taken for the stone to hit the object. 1

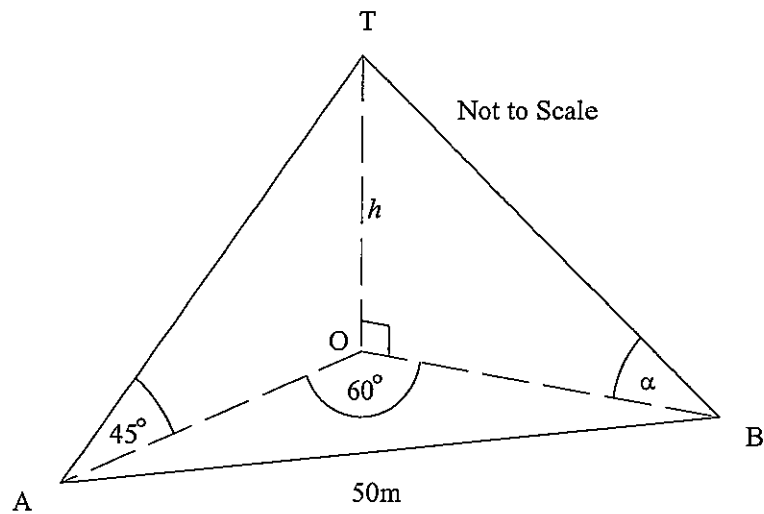
- (c) (i) Sketch the curve $y = x + \frac{4}{x}$ showing clearly all stationary points and asymptotes. 3

- (ii) Hence, find the values of k such that $x + \frac{4}{x} = k$ has no real roots. 1

Question 6 (12 marks) Use a **SEPARATE** Writing Booklet.

Marks

(a)



In the above diagram, the points A, B and O are in the same horizontal plane. A and B are 50 m apart and $\angle AOB = 60^\circ$. OT is a vertical tower of height h metres.

The angles of elevation of T from A and B respectively are 45° and α° (α is acute).

- | | | |
|-------|--|---|
| (i) | Explain why $AO = h$. | 1 |
| (ii) | Prove $h^2 \cot^2 \alpha - h^2 \cot \alpha + h^2 = 50^2$. | 2 |
| (iii) | Given that the tower is 30m high, find the angle α correct to the nearest degree. | 3 |

Question 6 - continued.

Marks

(b) (i) Verify that $\frac{d}{dx}\left(x \sin^{-1} x + \sqrt{1-x^2}\right) = \sin^{-1} x.$ **1**

(ii) Hence, using a similar expression, find a primitive of $\cos^{-1} x.$ **1**

(iii) The curves $y = \sin^{-1} x$ and $\cos^{-1} x$ intersect at $P.$ **1**

The curve $y = \cos^{-1} x$ also intersects the x axis at $Q.$

Show that P has co-ordinates $\left(\frac{1}{\sqrt{2}}, \frac{\pi}{4}\right).$

(iv) Find the area enclosed by the x – axis and the arcs OP and $PQ.$ **3**

Question 7(12 marks) Use a **SEPARATE** Writing Booklet.

Marks

- (a) A particle is moving in a straight line with acceleration given by

$$\frac{d^2x}{dt^2} = 9(x-2).$$

where x is the displacement in metres, from the origin O after t seconds.

Initially the particle is 4m to the right of O and it has a velocity of $V = 6\text{m/s}$.

- (i) Show that $V^2 = 9(x-2)^2$. **3**

- (ii) Find an expression for V and hence find x as a function of t . **3**

- (b) PQ is a variable chord of the parabola $x^2 = 4ay$ which subtends a right angle at the vertex.

- (i) If p and q are the parameters corresponding to the points P and Q, prove that $pq = -4$. **2**

- (ii) Show that the equation of the normal at P is
$$x + py = 2ap + ap^3.$$
 2

- (iii) Hence prove that the locus of the point of intersection of the normals at P and Q is the parabola $x^2 = 16a(y - 6a)$. **2**

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1

$$a) \int \frac{2}{\sqrt{36-x^2}} dx$$

$$= 2 \sin^{-1} \frac{x}{6} + C$$

1 r/w

$$b) \frac{d}{dx} (\tan^{-1}(\ln x))$$

$$= \frac{1/x}{1 + (\ln x)^2}$$

$$= \frac{1}{x + x(\ln x)^2}$$

1 r/w Subsequent errors ignored

$$c) A(-3, 8) \text{ and } B(7, -3) \quad m \quad n \quad 2:3$$

$$x = \frac{nx_1 + mx_2}{m+n}$$

$$= \frac{3 \times (-3) + 2 \times 7}{2+3}$$

$$= \frac{5}{5}$$

①

$$= 1$$

$$\text{Point is } (1, 18/5)$$

1 (1)

$$d) \frac{4}{x-2} \leq 2$$

C.P. $x=2$

$$\text{Consider } \frac{4}{x-2} = 2$$

$$2x-4=4$$

$$2x=8$$

$$x=4$$

Test

$$x=1$$

$$x=3$$

$$x=5$$

$$\frac{4}{1-2} \leq 2 \quad \text{True Testing}$$

$$\frac{4}{3-2} \leq 2 \quad \text{False}$$

$$\frac{4}{5-2} \leq 2 \quad \text{True}$$

$$\text{So in } x < 2 \text{ ① and } x \geq 4 \text{ ①}$$

$$f) u=x-3 \quad \int_4^5 \frac{x}{\sqrt{x-3}} dx$$

$$\text{When } x=5 \quad u=5-3$$

$$=2$$

$$x=4$$

$$u=4-3$$

$$=1$$

$$u=x-3$$

$$du=dx$$

$$= \int_1^2 \frac{u+3}{\sqrt{u}} du$$

$$= \int_1^2 u^{1/2} + 3u^{-1/2} du$$

$$= \left[\frac{2u^{3/2}}{3} + \frac{3u^{1/2}}{1/2} \right]_1^2$$

① Correct integral and limits

$$= \frac{2}{3} \times 2^{3/2} + 6\sqrt{2} - \frac{2}{3} - 6$$

① Correct integration

$$= \frac{2}{3} \times (\sqrt{2})^3 + 6\sqrt{2} - \frac{2}{3} - 6$$

② Correct substitution and attempt at tidying answer

$$= \frac{2}{3} \times 2\sqrt{2} + 6\sqrt{2} - \frac{20}{3}$$

$$= \frac{4}{3} \sqrt{2} + 6\sqrt{2} - \frac{20}{3}$$

$$= 22\sqrt{2} - 20 = 2(11\sqrt{2} - 10)$$

$$e) \quad 2x - y - 4 = 0 \quad y = mx + 3 \quad \theta = 45^\circ$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \quad y = 2x - 4$$

$$\therefore \tan 45 = \left| \frac{2 - m}{1 + 2m} \right|$$

$$\left| \frac{2 - m}{1 + 2m} \right| = 1 \quad (1)$$

$$\frac{2 - m}{1 + 2m} = 1 \quad \text{or} \quad \frac{2 - m}{1 + 2m} = -1$$

$$2 - m = 1 + 2m \quad \text{or} \quad 2 - m = -1 - 2m$$

$$3m = 1$$

$$m = \frac{1}{3}$$

(1)

$$3 = -m$$

$$m = -3$$

Question 2

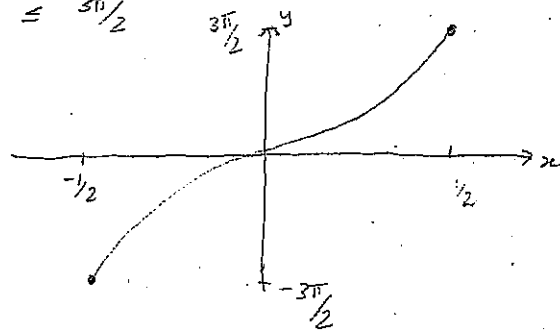
$$a) \quad f(x) = 3 \sin^{-1} 2x$$

$$-\frac{\pi}{2} \leq \frac{y}{3} \leq \frac{\pi}{2}$$

$$-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$$

$$-1 \leq 2x \leq 1$$

$$-\frac{1}{2} \leq x \leq \frac{1}{2}$$



(1) shape

(1) domain & range

$$b) \quad i) \quad \frac{d}{dx} (x \cos^2 x)$$

$$= x \times 2 \cos x (-\sin x) + \cos^2 x$$

$$= -x \sin 2x + \cos^2 x$$

(1) Correct product rule

(1) Correct derivative of $\cos^2 x$

$$ii) \quad \int x \sin 2x \, dx$$

Now

$$\int (\cos^2 x - x \sin 2x) \, dx = x \cos^2 x \quad \text{from (1)}$$

$$\therefore \int \cos^2 x - \int x \sin 2x \, dx = x \cos^2 x$$

$$\therefore \int x \sin 2x \, dx = \int \cos^2 x \, dx - x \cos^2 x \quad (1) \text{ for } \int \cos^2 x$$

$$= \int \frac{1 + \cos 2x}{2} \, dx - x \cos^2 x$$

$$= \frac{x}{2} + \frac{\sin 2x}{4} - x \cos^2 x + C$$

$$c) \quad P(x) = x^3 + ax^2 - 2x + b \quad x+1$$

$$P(-1) = 0 \quad \therefore (-1)^3 + a(-1)^2 - 2(-1) + b = 0$$

$$-1 + a + 2 + b = 0$$

$$a + b = -1 \quad (1)$$

$$P(3) = 4 \quad \therefore (3)^3 + a(3)^2 - 2(3) + b = 4$$

$$\therefore 27 + 9a - 6 + b = 4$$

$$9a + b = -17 \quad (2)$$

$$(2) - (1)$$

d) $f(x) = x - e^{-2x}$ $x = 0.3$

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ ① $f'(x) = 1 + 2e^{-2x}$

$\therefore x_1 = 0.3 - \frac{0.3 - e^{-0.6}}{1 + 2e^{-0.6}}$ ①-sub $f(0.3) = 0.3 - e^{-0.6}$
 $f'(0.3) = 1 + 2e^{-0.6}$

≈ 0.418615978

≈ 0.42 (2 dp) ①-correctly rounded answer

Question 3

a) $\lim_{x \rightarrow \infty} \frac{3x^2 + 1}{2 - 7x - 5x^2}$

$= \lim_{x \rightarrow \infty} \frac{3x^2/x^2 + 1/x^2}{2/x^2 - 7x/x^2 - 5x^2/x^2}$

$= \lim_{x \rightarrow \infty} \frac{3 + 1/x^2}{2/x^2 - 7/x - 5}$

$= -3/5$

② Correct with Marks Collect notation

① - Incorrect use of notation with collect answer
 Incorrect answer with minor error!

Since $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$

b) i) $\sqrt{3} \cos \theta - \sin \theta$ $r \cos(\theta + \alpha)$

$r \cos(\theta + \alpha) = r(\cos \theta \cos \alpha - \sin \theta \sin \alpha)$

$\sqrt{3} \cos \theta - \sin \theta = r(\cos \theta \cos \alpha - \sin \theta \sin \alpha)$

Equating coefficients

$r \cos \alpha = \sqrt{3}$

$r \sin \alpha = 1$

(DC)

Squaring and adding

$r^2 \cos^2 \alpha + r^2 \sin^2 \alpha = 3 + 1$

$r^2 = 4$

$r = 2, r > 0$

$2 \cos \alpha = \sqrt{3}$

$2 \sin \alpha = 1$

$\cos \alpha = \sqrt{3}/2$

$\sin \alpha = 1/2$

Since $0 < \alpha < \pi/2$

$\alpha = \pi/6$

①

①

$\sqrt{3} \cos \theta - \sin \theta = 2 \cos(\theta + \pi/6)$

ii) $2 \cos(\theta + \pi/6) = -1$

$\cos(\theta + \pi/6) = -1/2$

$\theta + \pi/6 = 2\pi/3, 4\pi/3$

$\theta = \pi/2, 7\pi/6$

② - correct angles and setting out

① - error, too many angles,

c) $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$ one incorrect angle

In $\triangle ADC$ $AC = 5$ by Pythagoras Th^m since $\triangle ADC$ is right angled.

In $\triangle ADB$ $AB = \sqrt{2^2 + 3^2}$

$= \sqrt{13}$ also by Pythagoras Th^m

Since $\triangle ACD$ is right angled

$\sin \alpha = 4/5$

$\cos \alpha = 3/5$

①

$\sin \beta = 3/5$

$\cos \beta = 4/5$

$$\sin(\alpha - \beta) = \frac{4}{5} \times \frac{3}{\sqrt{13}} - \frac{3}{5} \times \frac{2}{\sqrt{13}} \quad (1)$$

$$= \frac{12 - 6}{5\sqrt{13}}$$

$$= \frac{6}{5\sqrt{13}} \times \frac{\sqrt{13}}{\sqrt{13}}$$

$$= \frac{6\sqrt{13}}{65} \quad (1)$$

d) 1. $\angle WYX = \angle WZY$ Angle between tangent and chord equals the angle at the circumference in the alternate segment.

Similarly $\angle ZWY = \angle WXY$ (1) For both angle justifications.

$\angle WYZ = \angle XWY$ angle sum triangles (1)

WXY and WZY with two equal angles in both triangles

but $\angle WYZ$ and $\angle XWY$ are alternate and equal $\therefore WX \parallel YZ$ (1)

Question 4

a) $\int_0^{\ln 3} \frac{e^x}{e^x + 9} dx$

$$= \left[\ln(e^x + 9) \right]_0^{\ln 3} \quad (1)$$

$$= \ln(e^{\ln 3} + 9) - \ln(e^0 + 9)$$

$$= \ln 12 - \ln 10$$

$$= \ln 1.2 \quad (1)$$

b) $\binom{6}{r} 2^{6-r} x^{6-r} (-1)^r x^{-r}$

where $6 - 2r = 0$

$$r = 3 \quad (1)$$

$$\therefore {}^6C_3 8x^3 - \frac{1}{x^3}$$

$$= -160 \quad (1)$$

(c) Show statement is true for $p=1$

$$\therefore 4(1) + 3(1)^2 + 2(1)^3 = 9$$

which is div by 3

$\therefore$ statement is true for $p=1$ (1)

Assume statement is true for $n=k$

$$\therefore 4k + 3k^2 + 2k^3 = 3M \text{ where } M \text{ is a true integer}$$

RTP the statement is true for $p=k+1$

$$\begin{aligned}
 \text{Now } & 4(k+1) + 3(k+1)^2 + 2(k+1)^3 \\
 &= (k+1) (4 + 3(k+1) + 2(k+1)^2) \\
 &= (k+1) (4 + 3k + 3 + 2k^2 + 4k + 2) \\
 &= (k+1) (2k^2 + 7k + 9) \\
 &= 2k^3 + 7k^2 + 9k + 2k^2 + 7k + 9 \\
 &= 2k^3 + 9k^2 + 16k + 9 \quad \text{--- (1)} \\
 &= (2k^3 + 3k^2 + 9k) + 6k^2 + 12k + 9 \\
 &= 3M + 3(2k^2 + 6k + 3) \quad \text{--- from assumption} \\
 &= 3(M + 2k^2 + 6k + 3) \quad \text{--- (1)}
 \end{aligned}$$

where $M + 2k^2 + 6k + 3$ is integral.

If statement is true for $p=k$, then it is true for $p=k+1$. Since it is true for $p=1$, then true for $p=2$ & so on. It is true for all the integers p .

i) When $t=0$ $T = 100^\circ\text{C}$
 $t = 15$ $T = 70^\circ\text{C}$

$$100 = Ae^{k \times 0} + 22$$

$$100 = A + 22$$

$$A = 78$$

$$70 = 78e^{15k} + 22 \quad \text{--- (1)}$$

$$78e^{15k} = 48$$

$$e^{15k} = \frac{8}{13}$$

$$15k = \ln \frac{8}{13}$$

$$k = \frac{\ln \frac{8}{13}}{15}$$

e)

$$\begin{aligned}
 LHS &= \frac{\sqrt{1 + \sin 2\theta}}{\sqrt{1 - \sin 2\theta}} \\
 &= \sqrt{\frac{1 + 2\cos\theta\sin\theta}{1 - 2\cos\theta\sin\theta}} \div \cos^2\theta \\
 &= \sqrt{\frac{\sec^2\theta + \frac{2\cos\theta\sin\theta}{\cos 2\theta}}{\sec^2\theta - \frac{2\cos\theta\sin\theta}{\cos 2\theta}}} \\
 &= \sqrt{\frac{\sec^2\theta + 2\tan\theta}{\sec^2\theta - 2\tan\theta}} \quad \text{--- (1)} \\
 &= \sqrt{\frac{1 + \tan^2\theta + 2\tan\theta}{1 + \tan^2\theta - 2\tan\theta}} \\
 &= \sqrt{\frac{(1 + \tan\theta)^2}{(1 - \tan\theta)^2}} \quad \text{--- (1)} \\
 &= \frac{1 + \tan\theta}{1 - \tan\theta} \\
 &= RHS.
 \end{aligned}$$

Question 5

i) $v^2 = 27 + 18x - 9x^2$

$$\frac{1}{2} v^2 = \frac{1}{2} (27 + 18x - 9x^2)$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{d}{dx} \left\{ \frac{1}{2} (27 + 18x - 9x^2) \right\}$$

$$= 9 - \frac{9}{2} \times 2x$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 9 - 9x$$

$$= -9(x-1)$$

Since $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$$\ddot{x} = 9(1-x)$$

Since this is a multiple of the displacement

of the form $\ddot{x} = -n^2(x-a)$ the motion is S.H.

The centre of motion is 1 and the period is $\frac{2\pi}{3}$

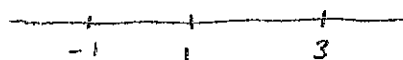
When $v = 0$

$$27 + 18x - 9x^2 = 0$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$x = 3 \text{ or } x = -1$$



Particle oscillates between -1 and 3

∴ amplitude is 2 (1)

ii) $\ddot{x} = -3^2(x-1)$

When $x = 2$ (1 cm from centre of motion)

$$\ddot{x} = -9(2-1)$$

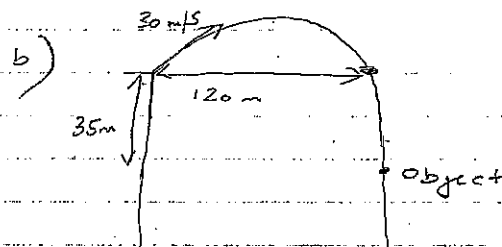
$$= -9 \text{ cm/s}^2$$

When $x = 0$

$$\ddot{x} = -9(-1)$$

$$= 9 \text{ cm/s}^2$$

(1) for both cases



$$x = 30t \cos \alpha \quad \text{--- (1)}$$

$$y = -5t^2 + 30t \sin \alpha \quad \text{--- (2)}$$

From (1) $t = \frac{x}{30 \cos \alpha}$ Sub into (2)

$$y = -5 \left(\frac{x}{30 \cos \alpha} \right)^2 + 30 \left(\frac{x}{30 \cos \alpha} \right) \sin \alpha \quad \text{--- (1)}$$

$$= \frac{-5x^2}{900 \cos^2 \alpha} + x \tan \alpha$$

$$y = \frac{-x^2}{180 \sec^2 \alpha} + x \tan \alpha$$

When $x = 120$ $y = -35$

$$-35 = \frac{-120^2}{180} \sec^2 \alpha + 120 \tan \alpha \quad \text{--- (1)}$$

$$-6300 = -14400(1 + \tan^2 \alpha) + 21600 \tan \alpha$$

$$14400 \tan^2 \alpha - 21600 \tan \alpha + 8100 = 0$$

$$48 \tan^2 \alpha - 72 \tan \alpha + 27 = 0$$

$$16 \tan^2 \alpha - 24 \tan \alpha + 9 = 0$$

$$(4 \tan \alpha - 3)^2 = 0$$

$$\therefore \tan \alpha = \frac{3}{4}$$

(1)

11) When $z = 120$ $\alpha = 36^\circ 52'$

$120 = 30 \pm \cos 36^\circ 52'$

$t = \frac{120}{30 \cos 36^\circ 52'}$

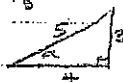
$= 5 \text{ seconds}$

or use

$\tan \alpha = \frac{3}{4}$

$\cos \alpha = \frac{4}{5}$

$\therefore t = \frac{4}{\cos \alpha} = 5$



(1)

c) 1) $y = x + \frac{4}{x}$

When $\frac{dy}{dx} = 0$

$\frac{dy}{dx} = 1 - \frac{4}{x^2}$

$1 - \frac{4}{x^2} = 0$

$\frac{d^2y}{dx^2} = \frac{8}{x^3}$

$\frac{4}{x^2} = 1$

$x^2 = 4$

$x = \pm 2$

When $x = 2$ $y = 2 + \frac{4}{2} = 4$

$x = -2$ $y = -2 - \frac{4}{2} = -4$

$= -4$

$\therefore$ sps are at $(2, 4)$ and $(-2, -4)$

When $x = 2$ $\frac{d^2y}{dx^2} = \frac{8}{2^3} > 0 \therefore (2, 4)$ is a

min s.p. When $x = -2$ $\frac{d^2y}{dx^2} = \frac{8}{(-2)^3} < 0 \therefore (-2, -4)$

is a max s.p.

When $\frac{d^2y}{dx^2} = 0$ $\frac{8}{x^3} = 0$ which has no solⁿs

$\therefore$ there are no inf pts.

$\lim_{x \rightarrow \infty} x + \frac{4}{x}$

$\lim_{x \rightarrow \infty} x + \frac{4}{x}$

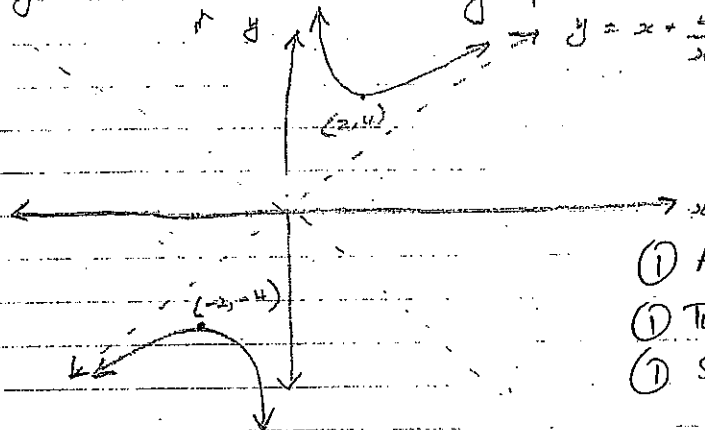
$= \infty$

$= \infty$

(∞, ∞)

$\therefore y = x$ is a skew asymptote

$y = x + \frac{4}{x}$



① Asymptotes

① Turning pts.

① Shape

$x = 0$ is a vertical asymptote

11) $x + \frac{4}{x} = k$ Since $y = x + \frac{4}{x}$ does not

exist when $-4 < x < 4$ $\therefore (2, 4)$ is a min

$(-2, -4)$ is a max

①

$x + \frac{4}{x} = k$ will have no solⁿ if

$-4 < k < 4$

Question 6

a) 1) In $\triangle AOT$ $\tan 45^\circ = \frac{h}{AO}$

$\therefore AO = \frac{h}{\tan 45^\circ}$ ①

But $\tan 45^\circ = 1 \therefore AO = h$

r/w

(0, 2)

ii) In $\triangle BOT$ $\tan \alpha = \frac{h}{OB}$

$\therefore OB = \frac{h}{\tan \alpha}$

$= h \cot \alpha$ ①

In $\triangle AOB$ by the cosine rule

$50^2 = OB^2 + OA^2 - 2 \times OB \times OA \cos 60$

$50^2 = h^2 \cot^2 \alpha + h^2 - 2h \cot \alpha \times h \times \frac{1}{2}$ ①

$50^2 = h^2 \cot^2 \alpha + h^2 - h^2 \cot \alpha$

iii) When $h = 30$

$50^2 = 30^2 \cot^2 \alpha + 30^2 - 30^2 \cot \alpha$

$900 \cot^2 \alpha - 900 \cot \alpha + 900 - 2500 = 0$

$900 \cot^2 \alpha - 900 \cot \alpha - 1600 = 0$

$9 \cot^2 \alpha - 9 \cot \alpha - 16 = 0$

$\cot \alpha = \frac{9 \pm \sqrt{81 - 4 \times 9 \times -16}}{18}$ ①

$= \frac{9 \pm \sqrt{657}}{18}$

* Since α is acute $\cot \alpha = \frac{9 + \sqrt{657}}{18}$ ①

$\tan \alpha = \frac{18}{9 + \sqrt{657}}$ ①

$\alpha = 27^\circ$ to nearest degree.

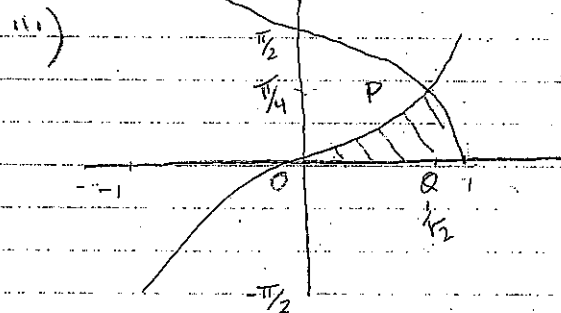
b) i) $\frac{d}{dx} (x \sin^{-1} x + (1-x^2)^{1/2})$

$= x \times \frac{1}{\sqrt{1-x^2}} + \sin^{-1} x + \frac{1}{2} (1-x^2)^{-1/2} \times -2x$ ①

$= \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x - \frac{x}{\sqrt{1-x^2}}$

$= \sin^{-1} x$

ii) $x \cos^{-1} x - \sqrt{1-x^2}$ is a primitive of ①



$\sin^{-1} x = \cos^{-1} x$

When $x = 1/\sqrt{2}$

$\sin^{-1}(1/\sqrt{2}) = \pi/4$

$\cos^{-1}(1/\sqrt{2}) = \pi/4$

$\sin^{-1} x = \cos^{-1} x$ ①

When $x = 1/\sqrt{2}$

iii) $\int_0^{1/\sqrt{2}} \sin^{-1} x \, dx + \int_{1/\sqrt{2}}^1 \cos^{-1} x \, dx$

Method 1

from i) and ii)

$A = \left[x \sin^{-1} x + \sqrt{1-x^2} \right]_0^{1/\sqrt{2}} + \left[x \cos^{-1} x - \sqrt{1-x^2} \right]_{1/\sqrt{2}}^1$

① $= \frac{1}{\sqrt{2}} \sin^{-1} \frac{1}{\sqrt{2}} + \sqrt{1 - (1/\sqrt{2})^2} - (0 \sin^{-1} 0 + \sqrt{1-0^2}) + 1 \cos^{-1} 1$

$- \sqrt{1 - (1/\sqrt{2})^2} - \left(\frac{1}{\sqrt{2}} \cos^{-1} \frac{1}{\sqrt{2}} - \sqrt{1 - (1/\sqrt{2})^2} \right) \sqrt{2} = 1$

$$= \frac{2}{\sqrt{2}} - 1$$

$$= \sqrt{2} - 1 \text{ sq units}$$

$$\text{OR } A = \int_0^{\pi/4} \cos y \, dy - \int_0^{\pi/4} \sin y \, dy \quad (1)$$

$$= [\sin y]_0^{\pi/4} - [-\cos y]_0^{\pi/4}$$

$$= \sin \frac{\pi}{4} - \sin 0 + \cos \frac{\pi}{4} - \cos 0 \quad (1)$$

$$= \frac{1}{\sqrt{2}} - 0 + \frac{1}{\sqrt{2}} - 1$$

$$= \frac{2}{\sqrt{2}} - 1$$

$$= \sqrt{2} - 1 \text{ sq units} \quad (1)$$

Question 7

$$a) \quad \ddot{x} = 9(x-2) \quad \text{When } t=0 \quad x=4 \quad \dot{x}=6$$

$$i) \quad \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 9(x-2)$$

$$\frac{1}{2} v^2 = \int 9(x-2) \, dx$$

$$\frac{1}{2} v^2 = \frac{9x^2}{2} - 18x + C$$

$$\text{When } x=4 \quad v=6$$

$$\therefore \frac{1}{2} \times 6^2 = \frac{9 \times 4^2}{2} - 18 \times 4 + C$$

$$18 = 72 - 72 + C$$

$$\therefore C = 18$$

$$\therefore \frac{1}{2} v^2 = \frac{9x^2}{2} - 18x + 18$$

$$v^2 = 9x^2 - 36x + 36$$

$$v^2 = 9(x^2 - 4x + 4)$$

$$v^2 = 9(x-2)^2$$

(p.16)

Q7

$$a) \quad \ddot{x} = 9(x-2)$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 9(x-2)$$

$$\frac{1}{2} v^2 = \int 9(x-2) \, dx$$

$$\frac{1}{2} v^2 = \frac{9}{2} (x-2)^2 + C$$

$$\text{When } x=4, \quad v=6$$

$$\frac{1}{2} \times 6^2 = \frac{9}{2} (4-2)^2 + C$$

$$18 = 18 + C$$

$$C = 0$$

$$\frac{1}{2} v^2 = \frac{9}{2} (x-2)^2$$

$$v^2 = 9(x-2)^2 \quad (3)$$

$$ii) \quad v = \pm 3(x-2)$$

$$\text{But when } x=4, \quad v=+6 > 0$$

$$\text{So } v = 3(x-2)$$

$$\frac{dx}{dt} = 3(x-2)$$

$$\frac{dx}{dx} = \frac{1}{3(x-2)}$$

$$t = \frac{1}{3} \int \frac{1}{x-2} \, dx$$

$$t = \frac{1}{3} \ln(x-2) + C$$

$$\text{When } t=0, \quad x=4$$

$$0 = \frac{1}{3} \ln 2 + C$$

$$C = -\frac{1}{3} \ln 2$$

$$t = \frac{1}{3} \ln(x-2) - \frac{1}{3} \ln 2$$

$$t = \frac{1}{3} \ln \left(\frac{x-2}{2} \right)$$

$$\frac{x-2}{2} = e^{3t}$$

$$x = 2e^{3t} + 2 \quad (3)$$

b) $x^2 = 4ay$
 Vertex = $(0, 0)$
 $P = (2ap, ap^2)$
 $Q = (2aq, aq^2)$

ii) $m_{PQ} = \frac{ap^2 - aq^2}{2ap - 2aq} = 0$

$= \frac{p}{2}$

$m_{QO} = \frac{q}{2}$

Since chord PQ subtends right angle at vertex,

$m_{PQ} \times m_{QO} = -1$

$\Rightarrow \frac{p}{2} \times \frac{q}{2} = -1$

$pq = -4$ (2)

iii) $x^2 = 4ay$

$y = \frac{x^2}{4a}$

$y' = \frac{x}{2a}$

$m_P = \frac{2ap}{2a} = p$

m of normal at P = $-\frac{1}{p}$

Eq. of normal at P

$\Rightarrow \frac{-1}{p} = \frac{y - ap^2}{x - 2ap}$

$-x + 2ap = py - ap^3$

$x + py = 2ap + ap^3$ (1)

iii) Similarly eq. of normal at Q

is $x + qy = 2aq + aq^3$ (2)

(1) - (2) $\Rightarrow (p - q)y = 2a(p - q) + a(p^3 - q^3)$

$y = 2a + a(p^2 + pq + q^2)$ (3)

Sub (3) into (1):

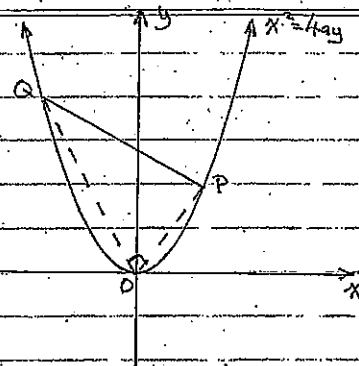
$x + 2ap + ap(p^2 + pq + q^2) = 2ap + ap^3$

$x + 2ap + ap^3 + ap^2q + apq^2 = 2ap + ap^3$

$x = -apq(p + q)$

Pt. of intersection of the normals at P & Q

$= (-apq(p + q), 2a + a(p^2 + pq + q^2))$



Since $pq = -4$ from (i):

$x = 4a(p + q) \Rightarrow p + q = \frac{x}{4a}$

$y = 2a + a(p^2 + 2pq + q^2 - pq)$

$= 6a + a(p + q)^2$

$y - 6a = a\left(\frac{x}{4a}\right)^2$

$y - 6a = \frac{x^2}{16a}$

$16a(y - 6a) = x^2$ (2)

(2)

(0.18)